
BioRTC Machine Learning for Neuroscience



Sussex AI



sussex
neuroscience



Alej & Dex & Paula AI

What will you learn?

Preprocessing, Analysing Data

Modelling + Predicting Patterns

Linear + Non-
Linear
Dimensionality
Reduction

VAEs

Classification

Regression

Linear Models

Artificial
Neural
Networks

Representing data (the basics)

When you learn to code we have these main data types:

Boolean, integer, float, string

In ML we typically work with numeric values.

Even language based modules use numeric representations of words such as word vectorization

Images (the basics)

Images are matrices

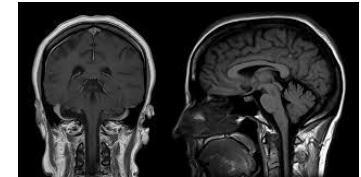
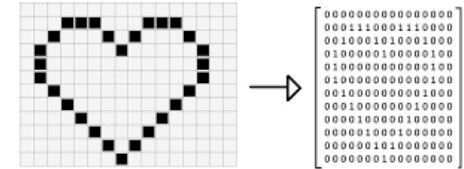
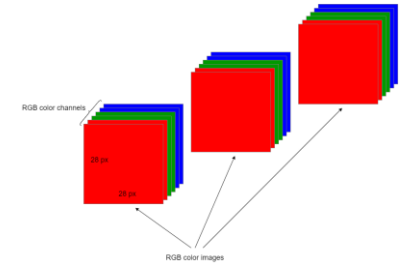
An image can be **represented numerically** as a matrix (or a multi-dimensional array)

Standard Images:

- Grayscale: $H \times W$: each pixel is 0–255 intensity; RGB color: $H \times W \times 3$: each pixel is (R, G, B), 0–255 per channel

Neuroscience Images:

- MRI – Structural brain scan ($X \times Y \times Z$)
- Histology / Microscopy – Tissue slices or volumes ($H \times W$ or $X \times Y \times Z$)
- Values are usually continuous (e.g. intensity, density), not just 0–255



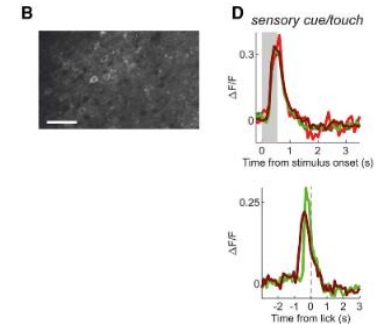
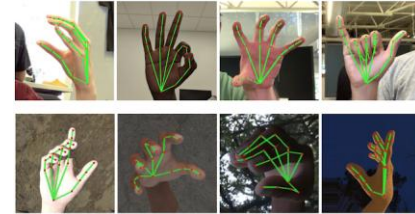
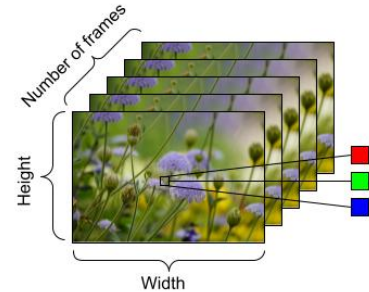
Videos/ Time-series data

Arrays of images

Can be treated as time-series data depending on the task

Neuroimaging Data (e.g. fMRI, calcium imaging):

- Also representable as matrices or tensors (multi-dimensional arrays).
- Common formats: fMRI: $4D \rightarrow X \times Y \times Z \times T$ (3D brain volumes over time)
- Calcium imaging: $3D \rightarrow H \times W \times T$ (2D field of view over time)
- Values are typically continuous (e.g., fluorescence intensity), not 0-255



Text/ strings

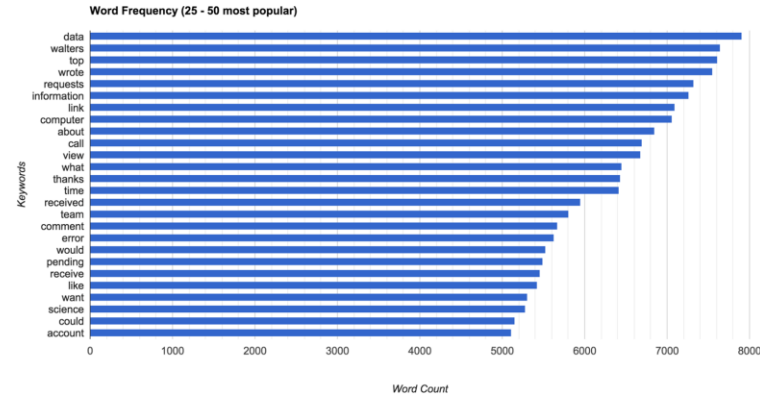
Much data is in the form of text
which is a complex problem

Classification with language can be
done in a number of ways

Naomi Surugaba [azlin@moa.gov.my]

Monday, March 10, 2014 1:18 PM

Dear Beloved Friend,
I know this message will come to you as surprised but permit me of my desire to go into business relationship with you.
I am Miss Naomi Surugaba a daughter to late Al-badari Surugaba of Libya whom was murdered during the recent civil war in Libya in March 2011, before his death my late father was a strong supporter and a member of late Moammar Gadhafi Government in Tripoli.
Meanwhile before the incident, my late Father came to Cotonou Benin republic with the sum of USD4, 200,000.00 (US\$4.2M) which he deposited in a Bank here in Cotonou Benin Republic West Africa for safe keeping.
I am here seeking for an avenue to transfer the fund to you in only you're reliable and trustworthy person to invest the fund. I am here in Benin Republic because of the death of my parent's and I want you to help me transfer the fund into your bank account for investment purpose.
Please I will offer you 20% of the total sum of USD4.2M for your assistance. Please I wish to transfer the fund urgently without delay into your account and also wish to relocate to your country due to the poor condition in Benin, as to enable me continue my education as I was a medical student before the sudden death of my parent's. Reply to my alternative email: missnaomisurugaba2@hotmail.com. Your immediate response would be appreciated.
Remain blessed,
Miss Naomi Surugaba.



Data sets

A data set is a collection of **related** data

In ML, models typically require the same number of inputs each time.

Image = (200,200) = 40,000 pixels

Model inputs: 40,000

Model outputs: 10 classes

Here are the classes in the dataset, as well as 10 random images from each:

airplane



automobile



bird



cat



deer



dog



frog



horse



ship



truck





Dimensionality Reduction and VAEs



Sussex AI

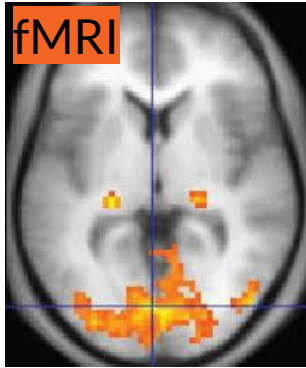


Contents

Why dimensionality reduction - why in CompNeuro?

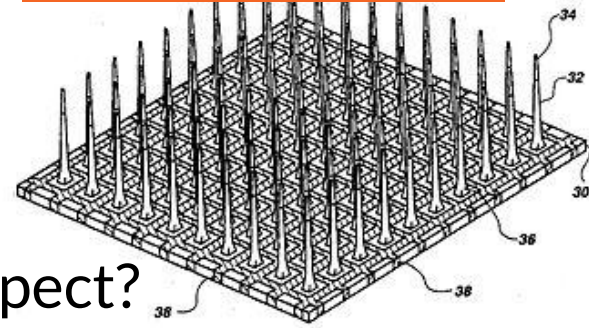
Methods:

- Linear Dimensionality Reduction - PCA
- Nonlinear Dimensionality Reduction - t-SNE, UMap, VAEs



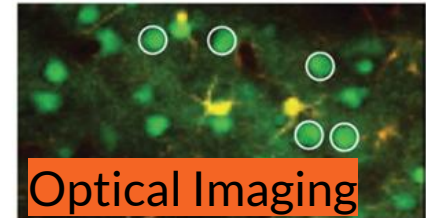
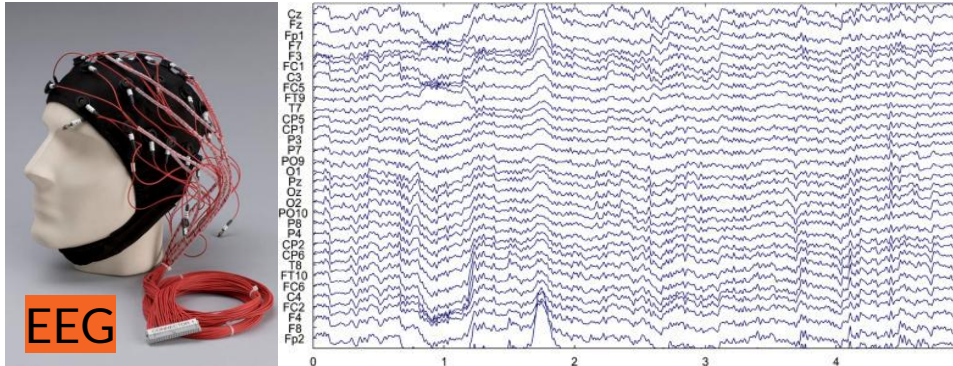
Why Dimensionality Reduction in Neuroscience?

Multi-Electrode Arrays

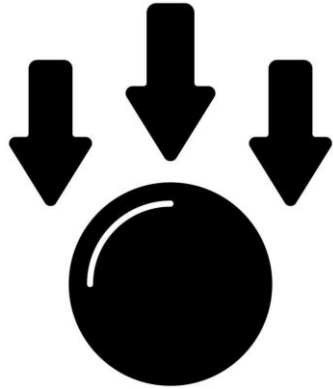


Multi- Dimensional Neural Data:

What is the most meaningful aspect?



Dimensionality reduction



VS

Volume of Ball
Starting Position
Physics of the world

Video of Ball falling:
Millions of pixel values

High-Dimensional
Explanation

Very few variables

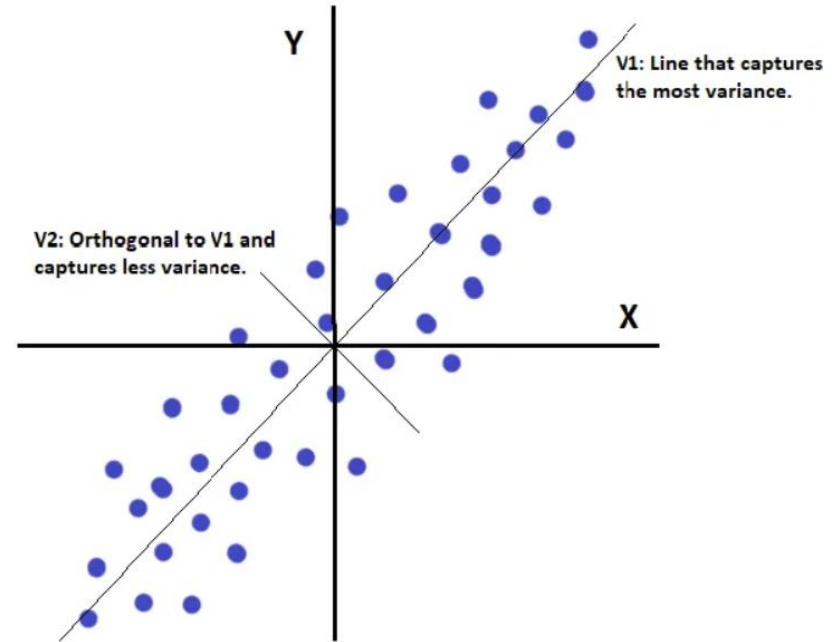
Low-Dimensional
Explanation

Principal component analysis

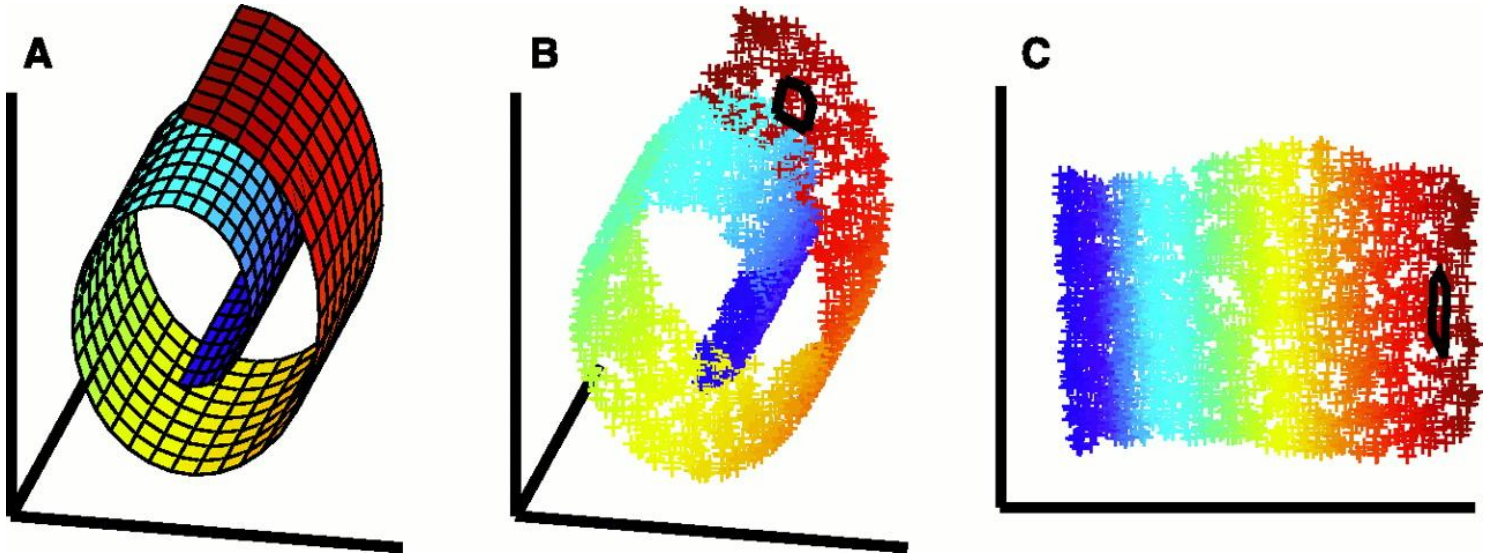
Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.

In simple terms it is reducing the number of variables from a data set - potentially at the cost of accuracy

This method helps in identifying the directions (or components) in which the data varies the most



Non-Linear Dimensionality Reduction

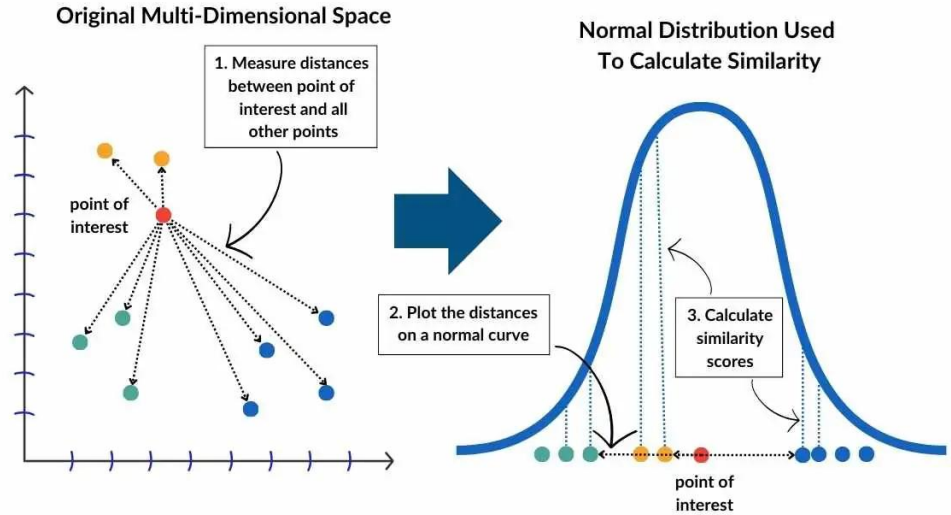


t-SNE

t-SNE is a nonlinear dimensionality reduction method used mainly for visualizing high-dimensional data in 2D or 3D.

It focuses on preserving local structure, meaning it keeps similar data points close together after reducing dimensions.

Unlike PCA, t-SNE does not create new axes as linear combinations. Instead, it builds a map that reflects the relative similarity between points, even if the original relationships are complex or nonlinear.



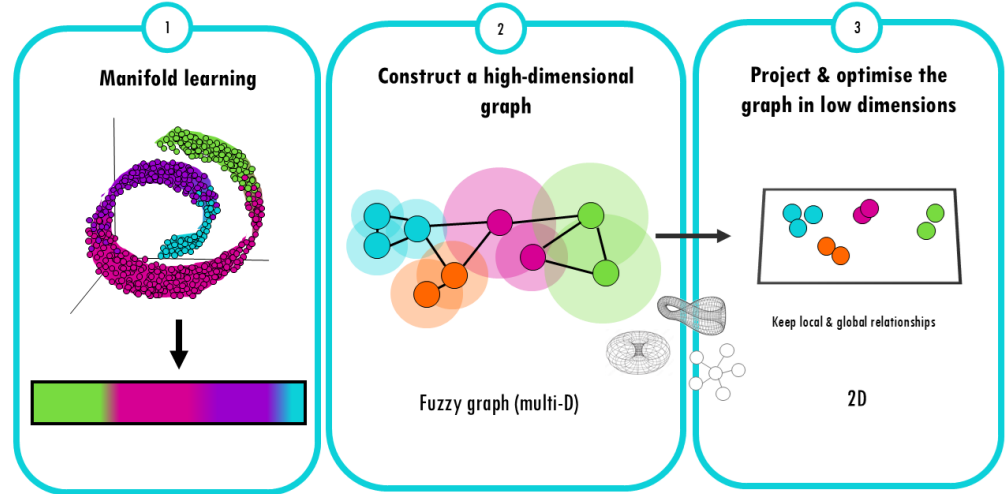
UMap

UMAP is a nonlinear dimensionality reduction method that, like t-SNE, is used to visualize high-dimensional data in 2D or 3D.

It aims to preserve both local and some global structure, making it better at maintaining the overall shape of the data.

Unlike PCA (linear) and t-SNE (local focus), UMAP builds a graph of neighborhood relationships and finds a low-dimensional space that best preserves those.

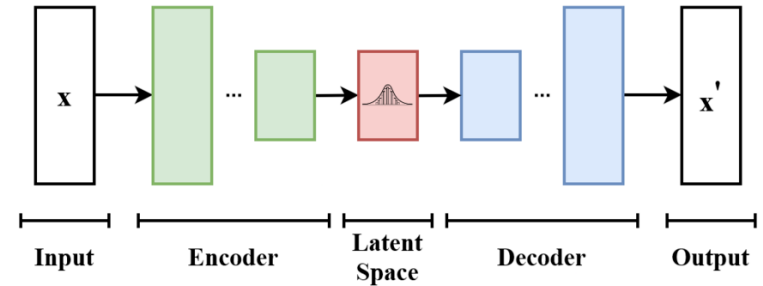
How does UMAP summarise many dimensions into 2?



Variational Autoencoders (VAEs)

A VAE is a deep learning-based nonlinear method for dimensionality reduction and generative modeling.

It learns to compress data into a low-dimensional latent space and then reconstruct it from that space - guided by probabilistic encoding (outputs are distributions, not fixed points).



Choosing a method

<i>Method</i>	<i>Type</i>	<i>Output</i>	<i>Best for</i>
PCA	Linear	Axes of variance	Denoising, compression
t-SNE	Nonlinear	3d/2d embeddings	Visualisation of local clusters
UMAP	Nonlinear	3d/2d embeddings	Visualisation + structural relationships
VAE	Deep learning, nonlinear	Low-dim latent space + decoder	Modelling, simulation, generative tasks

Problem 1:

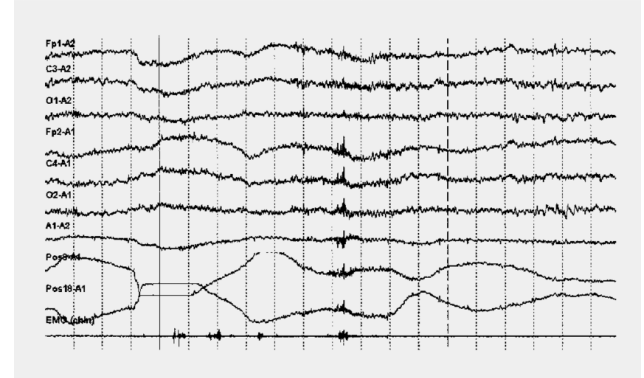
Removing artifacts from EEG recordings

You have collected EEG data for studying attention during a visual task. However, EEG signals have artifacts (e.g., eye movement) you need to clean before analysis. What method should you use?



Answer:

PCA, because it is a linear method that separates signals based on their variance. Since artifacts like eye blinks cause large variance, PCA can isolate and remove them, helping clean EEG data before analysis.



Problem 2:

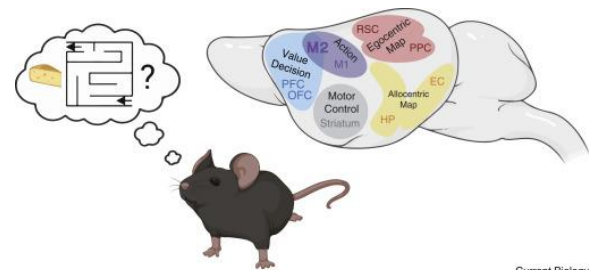
Spatial Navigation over Time

You are studying how the hippocampus encodes spatial memories in mice during maze navigation. You record from hundreds of neurons over time and want to reduce this complex activity into a low-dimensional space that captures the temporal dynamics of neural firing. You also want to reconstruct or simulate neural activity patterns to study memory replay.

What method should you use?

Answer:

VAE, because it learns a nonlinear, low-dimensional latent space that captures both the structure and variability of neural data. It allows for dimensionality reduction, while also providing a decoder to reconstruct or generate realistic activity patterns.



Current Biology

Labs

Worksheet

You will learn how to implement PCA, t-SNE, UMAP, and VAEs on a simple image dataset!

-> `dim_red.ipynb` on Github